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The Large-Scale Horizontal Structure of the Antarctic Circumpolar Current From FGGE Drifters

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The last decade of research in the Southern Ocean has shown that the Antarctic Circumpolar Current (ACC) is a complex system composed of narrow, high-speed currents separated by broad, quiescent zones. The circumpolar nature of this structure was examined using position and velocity data obtained from approximately 300 surface-drifting buoys deployed in the Southern Ocean during the First GARP Global Experiment (FGGE). The distribution of buoys on 1° × 1° squares shows that in some regions, most notably south of Australia, the buoys form three coherent bands of high buoy density which are separated by regions of low buoy density. The latitudes of these bands coincide with those of the Subtropical Front, Subantarctic Front, and Polar Front. To further examine the relationship between these fronts and buoy distribution, locations of the three fronts, determined from historical hydrographic data, were used to partition the buoys into zonal bands corresponding to front and nonfront regions. A mean buoy density and mean near-surface speed were then computed for each zonal band. High buoy densities were associated with all three fronts in the region south of Australia. Other regions also showed a tendency, although not as pronounced, for buoys to accumulate in fronts. The mean near-surface speeds suggest that the Subantarctic and Polar Fronts are circumpolar. Moreover, the mean near-surface speeds associated with the three frontal regions differ. Speeds within the Subantarctic and Polar Front regions are approximately twice that associated with the Subtropical Front.

1. INTRODUCTION

The past decade of research in the Southern Ocean has demonstrated that the Antarctic Circumpolar Current (ACC) is a complex system composed of narrow, high-speed currents separated by broad, quiescent zones. The most intense study of this zonation of the ACC took place in Drake Passage from 1975 to 1980 as part of the International Southern Ocean Studies program. Hydrographic and current measurements show that the ACC at this location is composed of two major fronts, characterized by large horizontal gradients of properties and large geostrophic shears, which separate three relatively uniform water mass zones [*Baker et al.*, 1977; *Nowlin et al.*, 1977; *Whitworth*, 1980; *Nowlin and Clifford*, 1982]. Ordered from north to south, these fronts and zones are Subantarctic Zone (SAZ), Subantarctic Front (SAF), Polar Frontal Zone (PFZ), Polar Front (PF), and Antarctic Zone (AAZ). In Drake Passage, average widths [*Nowlin and Clifford*, 1982] of the fronts and zones are 50–60 km and 200–300 km, respectively, and mean surface speeds within the frontal regions are 40–50 cm s^{-1} [*Whitworth et al.*, 1982]. Outside of Drake Passage the SAZ is bounded to the north by the Subtropical Front (STF) and the Subtropical Zone (STZ).

Regional studies at other locations have also described the meridional zonation of the ACC. *Heath* [1981] described the thermohaline structure and location of the STF and SAF south of New Zealand and found these fronts to be continuous over this region. *Emery* [1977], from historical hydrographic and expendable bathythermograph (XBT) data, showed that the SAF, PFZ, and PF are continuous over the area extending from south of Australia to east of Drake Passage. *Nowlin and Clifford* [1982] examined zonation of the ACC south of Australia and Africa, in addition to Drake Passage, and found analogous fronts to exist at all three locations.

The first study of the circumpolar nature of zonation of the ACC was that by *Clifford* [1982]. The results of this study, which are based on historical hydrographic and XBT data, suggest that the STF, SAF, and PF are continuous over the Southern Ocean and that they exist in the austral summer and winter. A unique opportunity to further examine the circumpolar nature of the fronts and zones associated with the ACC is provided by approximately 300 surface-drifting buoys (Figure 1) deployed in the Southern Ocean between 20°S and 65°S during the First GARP Global Experiment (FGGE). These buoys provided measurements of position, sea surface temperature, and sea level barometric pressure for a period of approximately 26 months, November 22, 1978, to January 13, 1981. Although the FGGE was primarily a meteorological experiment, oceanographers acquired a valuable data set that provides a synoptic view of the near-surface circulation of the Southern Ocean.

The objectives of this study were threefold. The first was to examine the meridional structure of the ACC using position and speed data obtained from the FGGE drifters in conjunction with the front locations determined from historical hydrographic data. The second study objective was to determine if the FGGE buoys were entrained into the high-speed currents associated with the frontal regions. A study of drifters in the eastern North Pacific [*Kirwan et al.*, 1978*b*] suggests that buoys have an affinity for regions of strong currents. The third objective was to determine the mean near-surface speeds within the fronts and zones over the ACC.

Section 2 describes the data used in this study. Discussions of the meridional distributions of the FGGE buoys and the relationship between buoy distributions and front locations are given in section 3. This section also presents a discussion of the mean near-surface speeds within the front and nonfront regions. Section 4 is a summary.

2. DATA

FGGE Drifters

The original drifting buoy data set consists of nonuniform time series of position, reported to the nearest hundredth of a degree of latitude and longitude, sea level pressure, and sea surface temperature. Discussions of the accuracy of these

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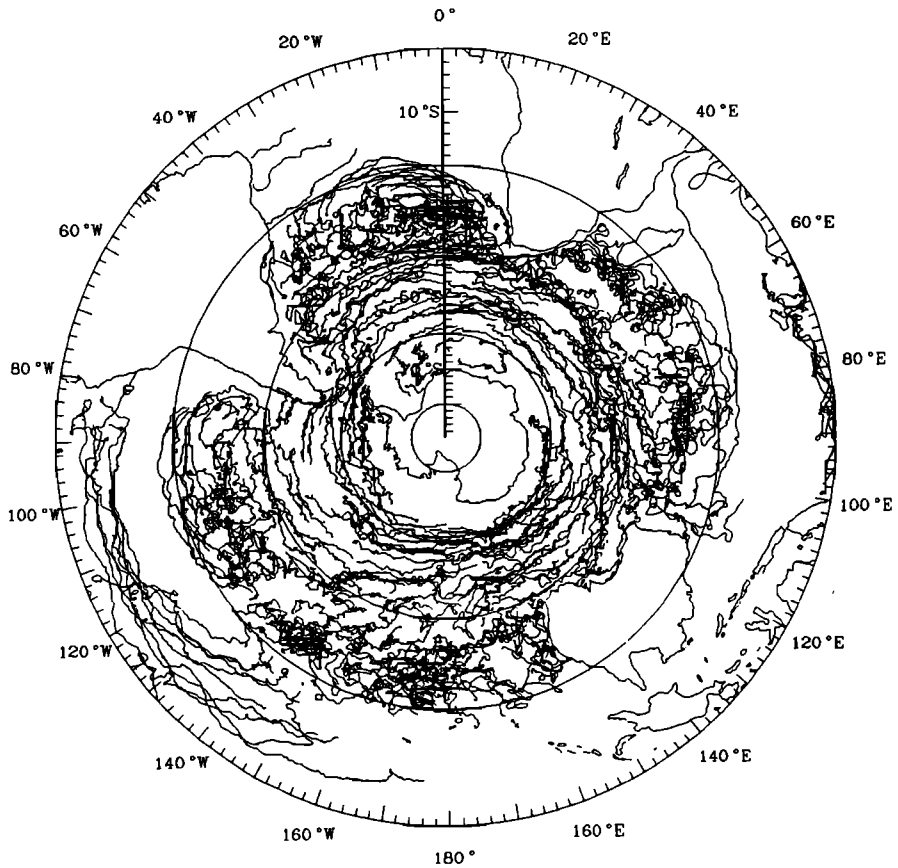

Fig. 1. Composite map of all the FGGE buoy trajectories in the Southern Hemisphere oceans.

measurements and the buoy data collection system are given by *Garrett* [1980*a*, *c*].

Before computing velocity time series, it was necessary to edit the original position series to remove spikes associated with random noise and to fill data gaps. The edited time series of position data were then smoothed and interpolated using a cubic spline smoothing routine [*Reinsch*, 1967]. The resulting smoothed data were resampled to construct a uniform time series of positions from which hourly velocities were computed using a 2-hour centered difference scheme. A detailed discussion of the buoy data reduction and processing techniques is given by *Patterson* [1985].

The hourly velocity data were used to compute mean east-west and north-south speed in 1° × 1° squares over the region 30°S to 70°S. One-degree squares were chosen because they provided sufficient buoy observations for calculation of average quantities while retaining the spatial resolution necessary to distinguish fronts. Also, the number of buoy tracks per degree of latitude in the region 30°S to 70°S was determined by counting the number of buoys drifting through each 1° × 1° square. Each buoy was counted only once in a particular square.

Approximately 39% of the drifters were equipped initially with drogues. No information is available as to whether the drogues remained attached for the entire lifetime of the buoys, although it is likely that they detached [*Meincke*, 1980]. A comparison of mean drift speeds for the main types of buoys [*Garrett*, 1980*b*, *c*] deployed in the Southern Ocean showed no obvious difference between drogued and undrogued buoys. A similar result was obtained for satellite-tracked buoys in the eastern North Pacific [*McNally*, 1981]. Moreover, there is no reliable method for correcting undrogued buoys for windage effects [*Kirwan et al.*, 1978*a*; *Peterson*, 1985]. In this study, no distinction has been made between drogued and undrogued buoys. Also, buoy velocities were not corrected for windage effects. For time scales of the order of days, wind effects on buoy velocity can be significant [*Peterson*, 1985]. However, when considering the large-scale mean circulation, the FGGE buoys provide a good representation of the interior flow [*Patterson*, 1985].

Front Locations

Most of the hydrographic and XBT data used by *Clifford* [1982] to determine front locations came from the Southern Ocean Atlas [*Gordon and Molinelli*, 1982]; however, other data sources were also included. From these data, 148 north-south transects, which crossed some or all of the fronts, were identified. Vertical sections of properties were constructed using hydrographic and XBT data from the stations along the transects. The location and width of individual fronts along each transect were determined from features in the property distributions or the horizontal density gradient. The specific criteria used to define and locate the fronts are given by *Clifford* [1982].

The circumpolar locations of the STF, SAF, and PF were determined for the austral summer and winter seasons. These positions showed little variability with season, thus the frontal locations are treated as a single data set in this study. These data provide 57 observations of the STF, 79 observations of the SAF, and 72 observations of the PF.

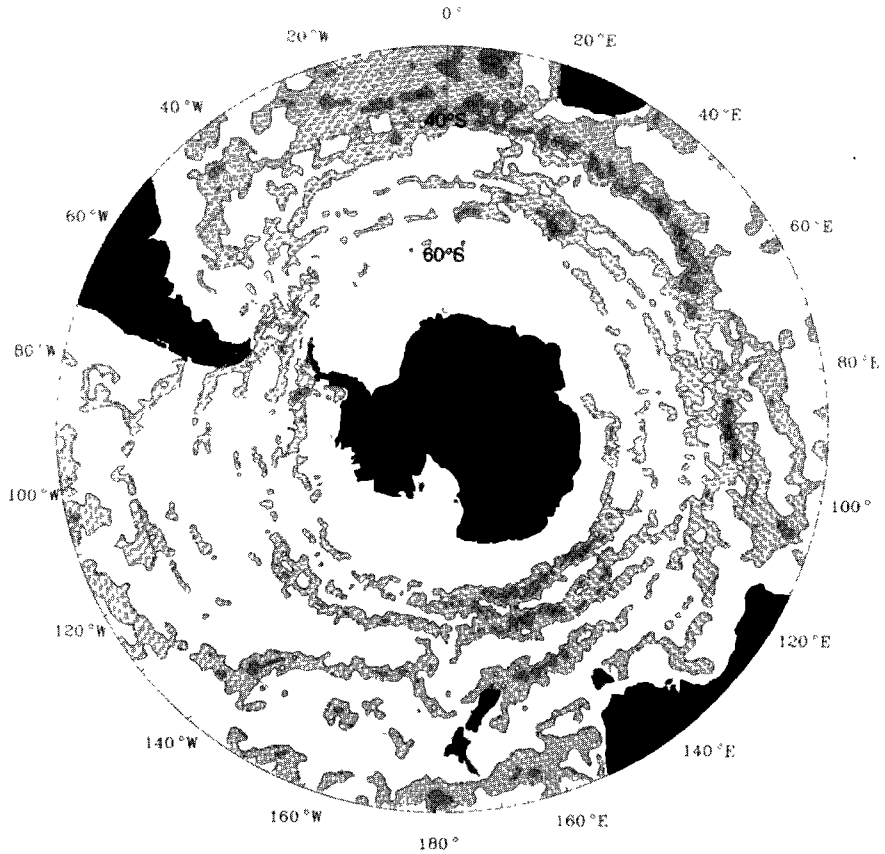

Fig. 2. Buoy tracks per degree of latitude over the region 30°S to 70°S. Light shading indicates two or more buoys; darker shading indicates four or more buoys.

3. DISCUSSION

Buoy Distributions

The distribution of buoy tracks on 1° squares over the region 30°S to 70°S is shown in Figure 2. In general, few 1° squares in the South Pacific and in the region south of 70°S were occupied by more than one buoy. By contrast, the South Atlantic, north of 35°S, shows a relatively high track count, reflecting the large number of buoys in the southern part of the South Atlantic Subtropical Gyre.

If it is assumed that the buoys accumulated in the frontal regions, then one might expect to see banded patterns in the buoy track distributions, i.e., regions of high track number separated by regions of low track number. In the region from 120°E to 170°E, south of Australia, such a banded pattern appears. Here three coherent bands of high track number are located at latitudes of 46°S to 49°S, 53°S to 56°S, and 57°S to 61°S, roughly the latitudes of the STF, SAF, and PF in this region [*Heath*, 1981; *Clifford*, 1982]. The maximum number of buoys in the bands, at 150°E, is four, five, and four, respectively; regions in between contain one or no buoys.

After passing New Zealand the northernmost band moves north and disappears between 175°E and 180°. It reappears at about 179°W and remains as an identifiable feature until 140°W, after which no continuous band is seen extending eastward toward the South American continent. The middle band shows a break near 176°E. Beyond 178°E it can be traced eastward to almost 120°W. The southernmost band ceases to appear as a continuous feature beyond 180°. However, there are regions of high buoy number south of 65°S extending eastward to 75°W, which give the impression of a continuous feature.

In Drake Passage, south of South America, two bands of high track number appear at latitudes of 56°S to 58°S and 60°S to 62°S. These latitudes correspond to the historical locations of the SAF and PF [*Whitworth*, 1980; *Clifford*, 1982], respectively. West of Drake Passage, between 95°W and 70°W, there is a region of high track number centered around 65°S. Upon entering Drake Passage, this band appears to merge with the region of high track number near 60°S. *Gordon and Molinelli* [1982] show that the 0° isotherm at 100 m (a reliable indicator of the PF) moves north about 7° of latitude (67°S to 60°S) between 90°W and Drake Passage. Thus the buoys may be reflecting the northward migration of the PF.

Eastward of Drake Passage there appears a broad region of high track number between 55°S and 60°S which extends to approximately 50°W. The SAF and PF in this region are in close proximity [*Clifford*, 1982], and averaging the buoy tracks over 1° squares may cause the two fronts to appear as a single feature.

From 50°W to the Greenwich Meridian there are no continuous bands of high track number extending across the South Atlantic. The few isolated areas of high track number that exist in this region are centered about 50°S and 54°S. At about 15°E, south of Africa, a banded pattern in the buoy track distribution is again apparent. The most distinct band occurs between 38°S and 41°S, approximately the latitudes spanned by both the Agulhas Return Current and the STF at this longitude [*Lutjeharms*, 1981]. Moving eastward, this band remains as a distinct feature over most of the Indian Ocean.

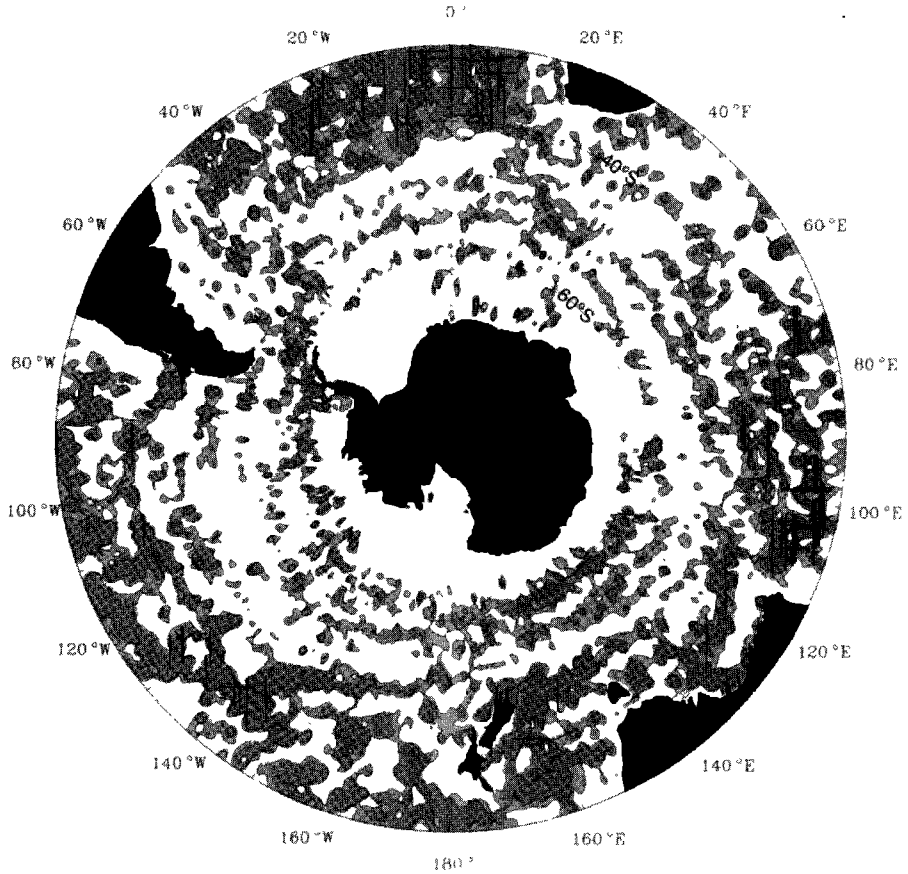

Fig. 3. Normalized buoy density × 100 over the region 30°S to 70°S. Light shading indicates values greater than 5; darker shading indicates values greater than 10.

A second region of high track number occurs south of Africa at about 48°S to 53°S, which are the approximate latitudes of the SAF and PF [*Clifford*, 1982]. Across the south Indian Ocean, there are no continuous bands south of 45°S, but isolated regions of high track number, centered around 52°S and 56°S, do occur.

The buoy track distribution gives the impression that the circulation of the ACC has a banded structure. Moreover, the bands occur, in some regions, at the latitudes of the STF, SAF, and PF. This would appear to be evidence for buoy accumulation in frontal regions. However, before concluding that this occurs, it is necessary to remove the effect of speed on buoy count, i.e., to distinguish between buoy count and buoy density. In a flow producing no change in an initially uniform buoy distribution, high-velocity regions will result in more buoy tracks in a given area. Thus, high track number does not necessarily indicate an affinity for a particular region. To correct for this effect, the buoy tracks in each 1° square were normalized by the mean speed in the 1° square. This calculation results in a buoy density distribution rather than a buoy track distribution.

The distribution of the normalized buoy density over the region 30°S to 70°S is shown in Figure 3. Again, in the region south of Australia, three coherent bands in buoy density are evident. As with the buoy tracks, these bands occur at the approximate latitudes of the STF, SAF, and PF. Eastward of Australia the bands of high buoy density are no longer continuous; however, there are numerous isolated regions of high buoy density which give the impression of continuous bands. The two regions of high buoy density in Drake Passage are located near 56°S and 61°S to 62°S, the approximate latitudes of the SAF and PF at this location. East of Drake Passage, at approximately 35°W, two bands of high buoy density, centered at 50°S and 54°S, appear. These bands can be traced across the South Atlantic to south of Africa and across the central Indian Ocean. Immediately south of Africa, at 38°S to 41°S, the buoy densities do not show a band analogous to that seen in the buoy track distribution. Rather, there are isolated regions of high buoy density, centered around 40°S, that extend eastward across the Indian Ocean.

One feature appears in the buoy density distribution that was not apparent in the buoy track distribution. In the South Atlantic, beginning at approximately 40°W, there is an additional band of high buoy density, centered around 60°S, that extends eastward to approximately 30°E, after which it turns south and then extends westward to the Greenwich Meridian. This band of high buoy density corresponds approximately to the historical location of the Weddell Gyre boundary [*Deacon*, 1979]. However, the property distributions used to delineate the boundaries of the Weddell Gyre indicate that it turns southward near 23°E [*Deacon*, 1979], as opposed to 30°E, which is suggested by the buoy density distribution.

Partitioning of Buoy Density and Speed Into Front and Nonfront Regions

The buoy distributions presented in Figures 2 and 3 show that the FGGE buoys were distributed in zonal bands. To test the correspondence between these bands and the locations of

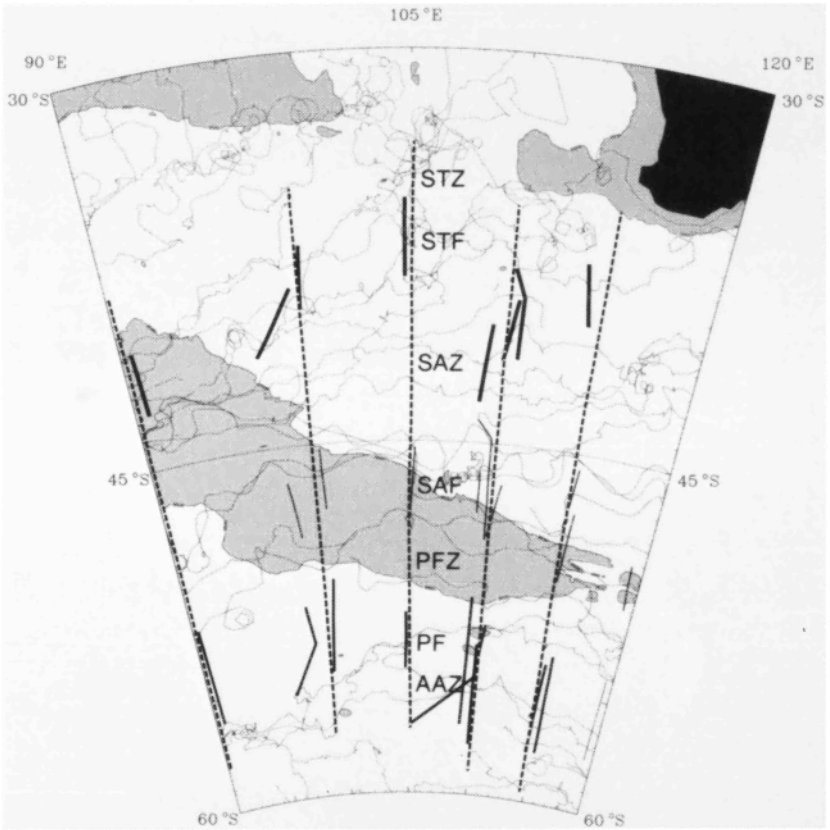

Fig. 4. Sector west of Australia. Line segments indicate the locations of the Subtropical Front (heavy lines), Subantarctic Front (light lines), and Polar Front (medium lines) determined by *Clifford* [1982]. Dashed lines are the longitudes along which the front locations were used to partition the buoys into zonal bands. The front and nonfront regions are identified along 105°E. Light shading indicates depths shallower than 3500 m. Dark shading is the southwest tip of Australia.

the STF, SAF, and PF, the historical frontal locations determined by *Clifford* [1982] were used to partition the buoys into zonal bands that correspond to front and nonfront regions. This then provides a way to compare buoy density and speed between zonal bands.

Beginning at the Greenwich Meridian, the region from 35°S to 70°S was divided into 12 sectors, each spanning 30° of longitude. Within each sector, longitude lines along which observations of the three fronts were available were identified, except for the Drake Passage sector in which only two of the fronts exist. The historical front locations were then used to divide the Southern Ocean at these longitudes into zonal bands corresponding to front and nonfront regions. Since it was not possible to delineate the northern boundary of the STZ and the southern boundary of the AAZ, these regions were taken to extend approximately 3°–4° of latitude north and south of the STF and PF, respectively. Therefore these zones may not truly represent the STZ and AAZ.

The front locations represent single observations of features that exhibit spatial variability. Therefore, where possible, two or three observations of individual fronts were averaged to obtain the front location. In all there were 39 lines along which the fronts could be identified. However, the number of lines varied from sector to sector. Two sectors (31°E to 60°E and 119°W to 90°W) have only two lines, whereas others have as many as five. In one sector, 29°W to 0°, the historical front locations provided only a single observation of all three fronts. A representative sector is shown in Figure 4.

An implicit assumption in this approach is that the initial FGGE buoy distribution was uniform, i.e., the buoys were not all deployed in fronts or in the regions between fronts. The buoy deployment positions (Figure 5) indicate that the buoys were, initially, more or less uniformly distributed over the Southern Ocean. An objective of the FGGE was to have a nearly uniform buoy array that would provide a nominal spatial resolution of 1000 km; no point in the ocean was to be more than 500 km from a buoy. Buoy deployment took place over a period of approximately six months, with the maximum buoy coverage occurring in late May 1979. At that time, 80% of the ocean from 20°S to 65°S was within 500 km of an operational buoy [*Fleming et al.*, 1979].

Once the buoys were partitioned into zonal bands, the number of buoy tracks in each zone was computed. Because the width of a zone may vary from line to line, the buoy track counts were normalized by the width of the zone to give the average number of buoy tracks per degree of latitude. This value was then normalized by the mean speed of the zone to give a buoy density for that zone. The buoy densities and near-surface speeds thus obtained were averaged to give a global mean value for each zone and a mean value for each zone within individual sectors. The results of these calculations are discussed in the following section.

Mean Buoy Densities and Near-Surface Speeds

The global mean buoy densities and near-surface speeds for individual zones are shown in Table 1. Mean buoy densities associated with the STZ and STF are essentially equal and approximately twice the value of the buoy densities associated

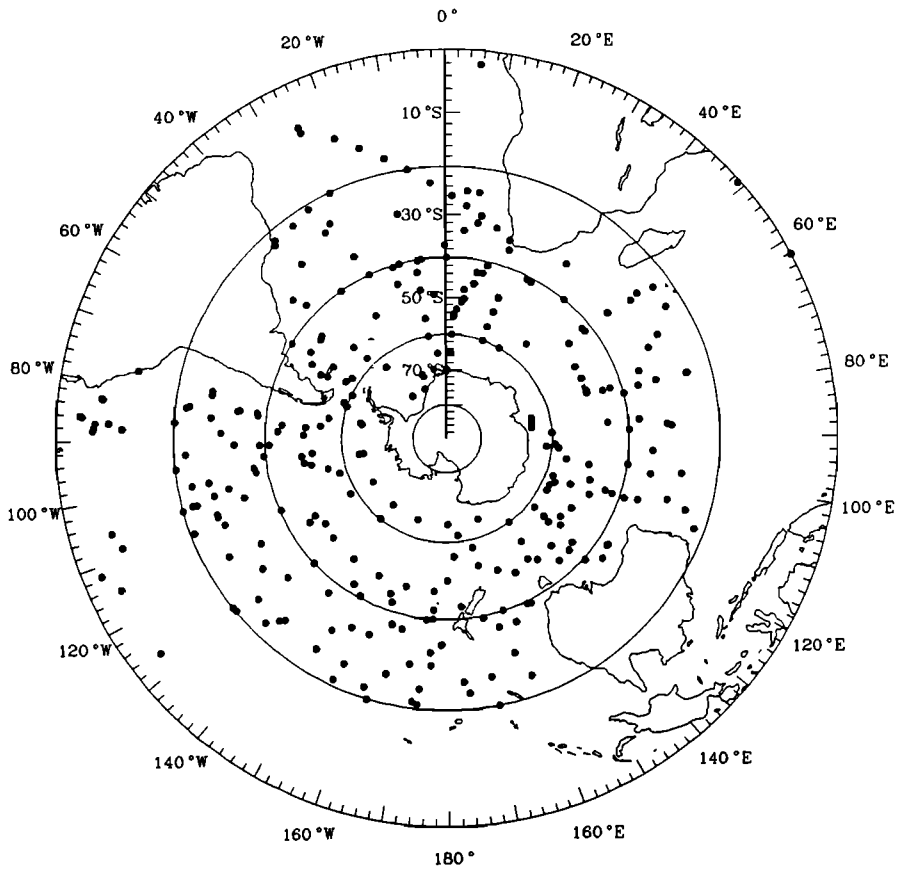

Fig. 5. Deployment positions of the FGGE drifting buoys.

with the other zones. Pairwise comparison of the buoy densities (Duncan's multiple range test, $\alpha = .05$) shows that the buoy densities of these two zones are statistically different from those of the remaining zones. The higher buoy densities associated with the STZ and STF probably reflect the longer lifetime of the buoys at these latitudes [*Patterson*, 1985]. Of the remaining zones the highest buoy density is associated with the PF. However, the buoy density associated with the PF is not statistically significantly ($\alpha = .05$) different from that of the PFZ and AAZ.

The highest mean near-surface speeds are associated with the SAF and PF regions. The mean speed within these frontal zones is approximately 40 cm s^{-1}, which is almost twice that associated with the STF. The mean near-surface speeds obtained for the SAF and PF regions are in agreement with measured speeds of the currents associated with these fronts in Drake Passage [*Whitworth et al.*, 1982]. Mean speeds of the nonfront regions range from 23 to 35 cm s^{-1}, with the highest speed associated with the PFZ. Pairwise comparison shows that the speeds of the SAF and PF are statistically different ($\alpha = .05$) from those of the nonfront regions to either side. Mean speeds of the four nonfront regions are not different from one another or from that associated with the STF.

TABLE 1. Mean Buoy Density × 100 and Near-Surface Speed Computed for Individual Zones From the Entire Data Set

	STZ	STF	SAZ	SAF	PFZ	PF	AAZ
Buoy density	7.3	8.3	4.4	4.7	3.9	5.9	3.6
Near-surface speed	23.2	25.5	27.0	42.7	34.7	40.1	28.2
Number of observations	35	35	39	39	39	39	39

The number of observations used for each zone is shown. Speed values are centimeters per second.

Mean buoy densities for each zone in the individual sectors are given in Table 2. In general, the buoy densities associated with the STZ and STF are higher than those associated with the other zones. All but four sectors show higher buoy densities associated with the STF than with the nonfront regions to either side. The highest buoy density associated with the SAF is found in sector 4. In fact, over the region 61°E to 180°, buoy densities of the SAF are higher than those of the zones to either side. The highest buoy densities associated with the PF are found in sectors 1, 5, and 10. Only one sector, sector 6, shows higher buoy densities associated with all three fronts.

The mean near-surface speeds associated with the zones in the individual sectors (Table 3) indicate that in general the speed of the STF region is lower than those associated with the SAF and PF regions. The highest speeds (relative to the global mean) associated with the STF occur in the sectors south of Africa (sectors 1 and 2) and in the sector east of South America (sector 11). The lowest speeds occur in the central Pacific sectors (sectors 7 to 9). The highest speeds associated with the SAF and PF are found in sectors 2, 3, and 6. Across the central Pacific the mean speeds within these two frontal regions are lower than their global means.

The speeds given in Table 3 can be used to investigate the circumpolar nature of the fronts. The speeds within the zones that are circumpolar (all but STZ and STF) show a distinct meridional zonation, with higher speeds associated with the frontal regions, in all but three sectors. The banding in the

TABLE 2. Normalized Buoy Density × 100 for Each Zone in the Individual Sectors

	Sector											
	1 1°– 30°E	2 31°– 60°E	3 61°– 90°E	4 91°– 120°E	5 121°– 150°E	6 151°E– 180°	7 179°– 150°W	8 149°– 120°W	9 119°– 90°W	10 89°– 60°W	11 59°– 30°W	12 29°W– 0
STZ	10.0	4.0	3.5	8.6	9.0	6.5	10.9	13.2	4.4	—	6.8	4.2
STF	5.3	9.0	4.2	10.3	5.4	9.4	11.5	19.6	7.3	—	3.5	5.2
SAZ	4.2	0.7	3.7	3.8	3.6	4.0	4.6	1.7	1.6	3.3	3.7	17.4
SAF	2.9	1.3	5.7	7.3	5.7	6.1	4.6	3.2	3.5	5.5	6.5	1.0
PFZ	4.1	2.4	3.1	3.5	5.6	5.2	4.1	4.6	1.3	6.4	2.6	3.5
PF	9.6	4.8	3.2	3.0	11.5	7.1	2.2	4.9	3.4	15.6	3.5	1.7
AAZ	2.7	3.5	1.4	3.0	5.6	0.7	1.5	4.5	5.0	9.5	4.1	1.4
Number of observations	3	2	3	5	5	4	4	3	2	4	3	1

The number of observations available for each zone in the sectors is shown. Values for sector 12 are based on one observation and do not represent means.

speeds is most evident over the region 1°E to 180°. In sectors 7, 9, and 10 the higher speed within the PFZ probably results from discrepancies between the historical locations of the PF and SAF and the location detected by the buoys. In particular, this may be true for sector 10, which includes Drake Passage. The estimated speed within the PF region in this sector is 25 cm s^{-1}. Direct measurements indicate speeds within the PF to be almost twice this value. The difference between the speed estimated from the buoys and the actual speed may be the result of the mesoscale variability associated with the PF. The PF at Drake Passage meanders [*Legeckis*, 1977] and sheds cold-core rings into the PFZ [*Joyce and Patterson*, 1977; *Joyce et al.*, 1981; *Peterson et al.*, 1982; *Hofmann and Whitworth*, 1985]. Additionally, the PF undergoes north-south migrations of approximately 100 km [*Klinck*, 1985; *Hofmann and Whitworth*, 1985]. The historical front positions used to determine the boundaries of the PF and PFZ in Drake Passage do not account for these processes. Therefore buoys that were actually in the PF may have been counted in the PFZ, thereby producing a PF speed that is too low and a PFZ speed that is too high.

Inclusion of the STZ and STF still shows meridional zonation in speed in sectors 1 and 2, which encompass the area south of Africa, sector 6, which is the region south of Australia, and sector 11, which is the region eastward of South America. Sectors 1 and 11 are regions in which the northern boundary of the ACC comes into contact with the Agulhas Return Current and the Brazil Current, respectively. In sectors 2 and 6 the STF flows over large bathymetric features (discussed in following sections). Elsewhere, the speed within the STF is lower than that of the SAZ. Across the central Pacific there is no sharp distinction between the speeds associated with the STF and those of STZ. The lack of a clear indication of the STF in the speed values in this region may be because the buoys missed the front, the historical front positions do not adequately represent the location of the STF, or the STF is not a well-defined feature over this region.

Comparison of Buoy Tracks and Front Locations

The buoy densities and near-surface speeds presented in Tables 2 and 3 indicate that differences between front and nonfront values are more pronounced in some sectors. Sector 6, which encompasses the region south of Australia, is one such example. A possible reason for this is suggested when the buoy tracks and historical front locations are compared with the bathymetry in the individual sectors. From 90°E to 135°E (Figure 6*a*) the STF locations and buoy tracks exhibit considerable variability in latitude. However, eastward of the Tasman Plateau (near 145°E) the latitudinal variability in the STF locations and buoy tracks decreases. As the buoys and STF flow across the Tasman Plateau and toward the Campbell Plateau (near 170°E), they remain near a latitude of approximately 48°S. Eastward of the Campbell Plateau (Figure 6*b*) the STF and buoys drift to the north, although the drifter trajectories are not coincidental with the STF locations. There

TABLE 3. Mean Near-Surface Speed in Centimeters per Second for Each Zone in the Individual Sectors

	Sector											
	1 1°– 30°E	2 31°– 60°E	3 61°– 90°E	4 91°– 120°E	5 121°– 150°E	6 151°E– 180°	7 179°– 150°W	8 149°– 120°W	9 119°– 90°W	10 89°– 60°W	11 59°– 30°W	12 29°W– 0
STZ	39.8	45.3	30.8	18.1	19.4	17.1	10.1	9.0	9.3	—	35.1	21.7
STF	41.9	51.2	33.8	19.0	21.2	22.8	9.0	9.0	10.6	—	45.2	25.7
SAZ	30.6	34.1	48.6	27.2	34.1	20.9	17.0	19.5	21.1	27.3	30.5	13.4
SAF	36.1	70.4	51.4	40.0	38.7	73.5	33.8	34.8	21.9	44.5	34.4	33.1
PFZ	21.3	34.9	39.5	39.1	30.5	38.0	31.5	30.7	29.3	37.5	31.3	33.4
PF	40.9	52.0	50.9	43.1	34.5	43.1	27.8	34.5	30.9	25.4	40.1	57.5
AAZ	22.7	35.7	33.2	33.1	23.3	32.6	28.9	31.9	19.1	30.4	27.9	20.0

The number of observations available for each zone in the individual sectors is the same as in Table 2. Values for sector 12 are based on one observation and do not represent means.

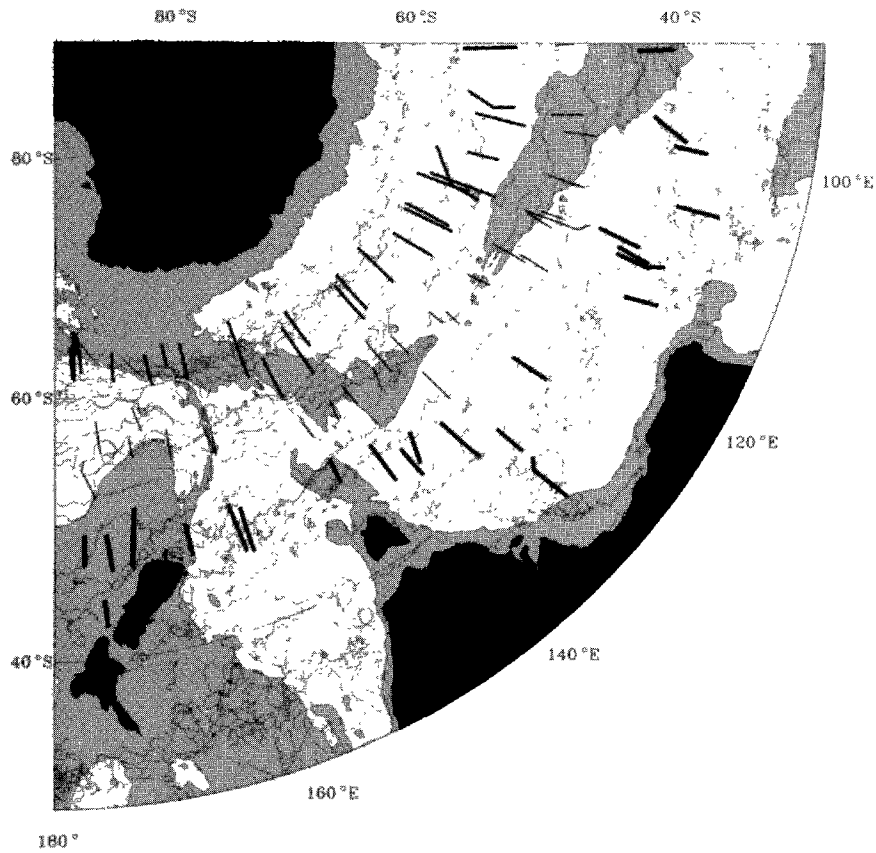

Fig. 6*a*. Buoy tracks in the region 90°E to 180°. Line segments are the locations of the Subtropical Front (heavy lines), Subantarctic Front (light lines), and Polar Front (medium lines) determined by *Clifford* [1982]. Light shading indicates depths shallower than 3500 m.

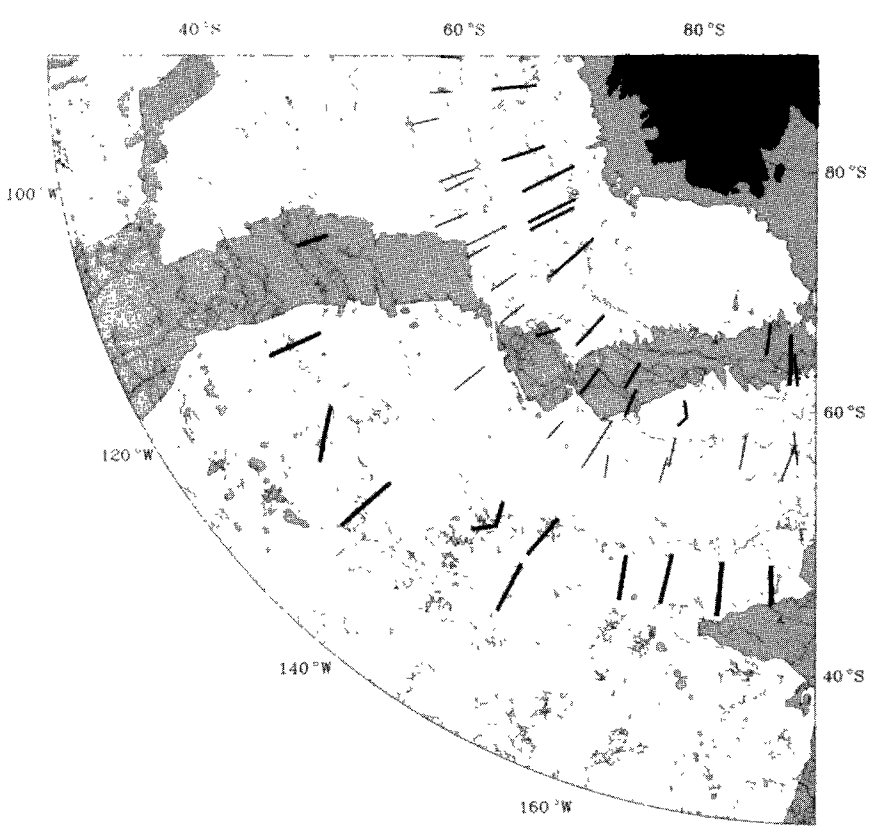

Fig. 6*b*. Same as 6*a* except for the region extending from 180° to 90°W.

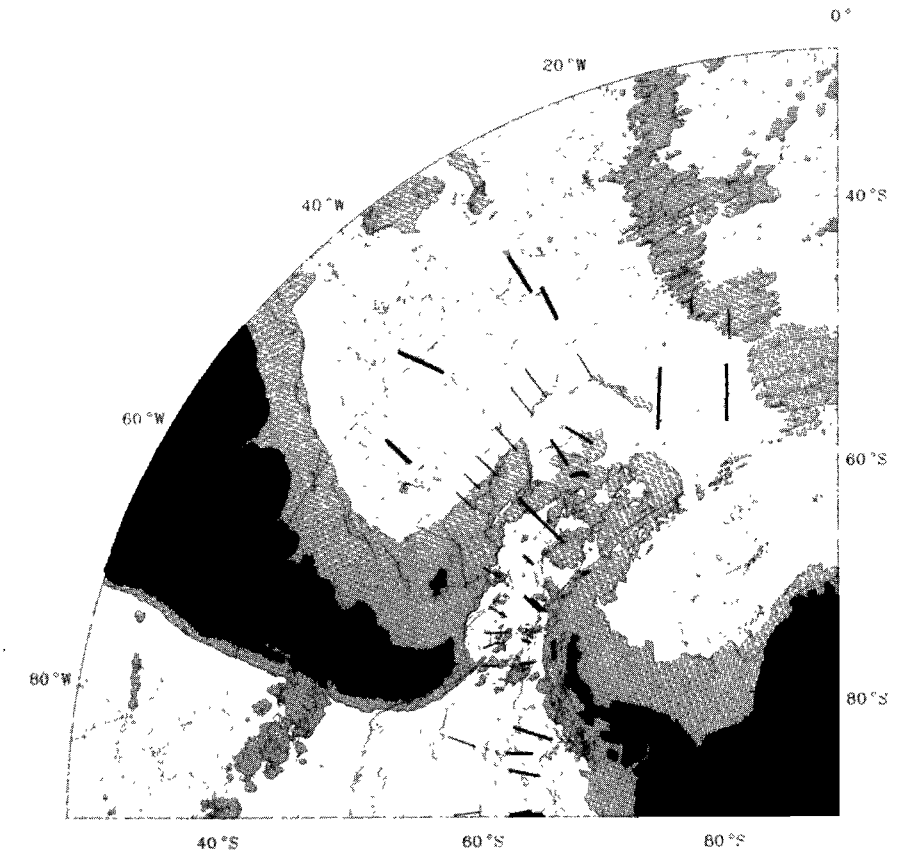

Fig. 6*c*. Same as 6*a* except for the region extending from 90°W to the Greenwich Meridian.

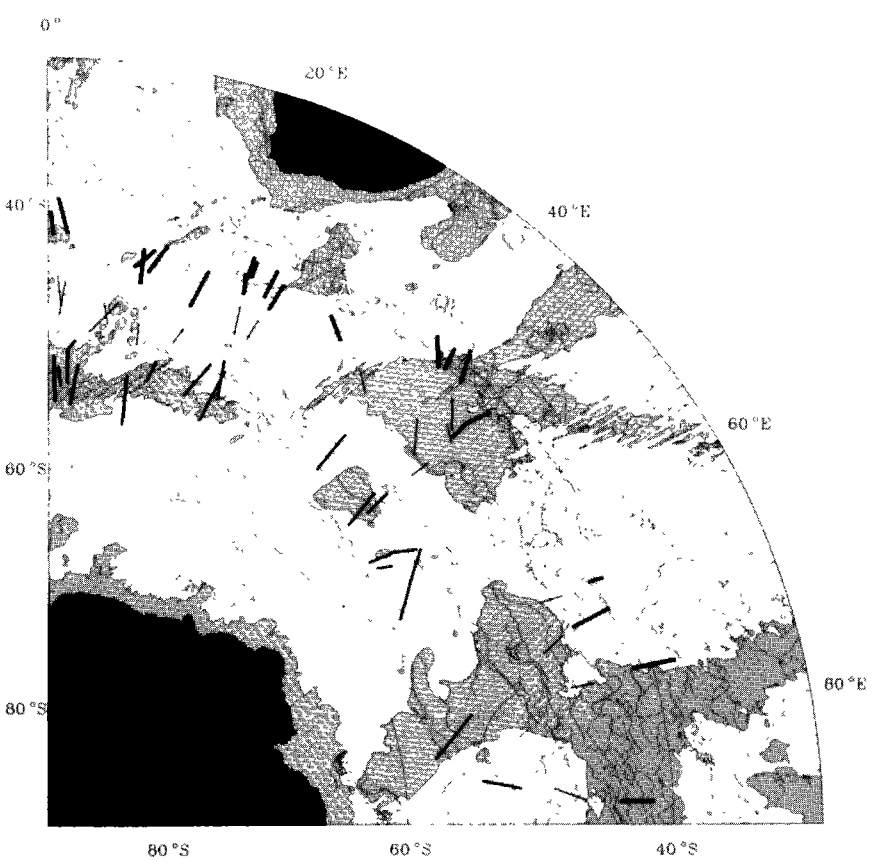

Fig. 6*d*. Same as 6*a* except for the region extending from the Greenwich Meridian to 90°E.

is no obvious agreement between buoy tracks and STF locations across the central Pacific.

There is relatively good correspondence between the buoy tracks and SAF locations over the region south of Australia (Figure 6*a*). Note that in this area the SAF flows along and over the Indian-Antarctic Rise, over the Macquarie Ridge (near 158°E), and around the tip of the Campbell Plateau. Similarly, the PF locations and buoy tracks show the best agreement where the PF flows over and along the Indian-Antarctic Rise. *Heath* [1981] suggested that the locations of these three fronts were controlled by the bathymetry in this region.

Between the Indian-Antarctic Rise and Macquarie Ridge, at 52°S to 56°S, the buoy tacks trace a large-amplitude wave. This feature appears in all the tracks, which suggests that there may be a permanent standing wave between these two bathymetric features. A similar though less pronounced wave pattern is seen at 58°S–60°S.

In the region of the East Pacific Rise (Figure 6*b*) the buoy tracks and SAF locations tend to align along latitudes of 57°S to 58°S. However, as reflected by the mean buoy densities and near-surface speeds, the distinction here between front and nonfront regions is not as pronounced as that south of Australia. To the east of South America (Figure 6*c*) along the northern edge of the Falkland Plateau there is again good agreement between the buoy tracks and SAF locations.

From 30°W to the Greenwich Meridian the front locations are not defined; therefore it is not possible to comment on the effect of the Mid-Atlantic Ridge. However, eastward of the Mid-Atlantic Ridge, at 1°E to 2°E, there is a correspondence between the buoy tracks and PF locations which persists over much of the region south of Africa.

The buoy tracks in the region of the Crozet Plateau (near 45°E; Figure 6*d*) indicate that the buoys are either confined to the region of the STF near 40°S or confined to the SAF along the southern edge of the Plateau. To the east of the Crozet Plateau the buoy tracks indicate considerable meandering. This region is known to have a high incidence of mesoscale (100–300 km) disturbances [*Lutjeharms and Baker*, 1979; *Colton and Chase*, 1983; *Patterson*, 1985].

The above comparisons suggest that the location of a front may not exhibit much spatial variability in regions where it flows over large bathymetric features. Hence, in these areas, partitioning of the buoys on the basis of historical front locations will give a fairly accurate representation of the number of buoys in front and nonfront regions. In other regions, such as the central Pacific, the fronts may exhibit considerable spatial variability. Therefore, the historical front location may not reflect that observed by the buoys.

4. SUMMARY

The buoy distributions presented in Figures 2 and 3 show that the FGGE buoys tended to accumulate in regions which correspond to the historical locations of the STF, SAF, and PF. The implication is that the fronts are regions of horizontal flow convergence at the surface. The best correspondence between the historical front locations and drifter tracks is found in areas where the fronts flow over large bathymetric features. These regions also show the clearest indication of a banded structure and the strongest meridional zonation in speed. These observations suggest that the bathymetry in these regions may inhibit the lateral motion of the fronts.

The fact that the FGGE buoys appeared to have an affinity for frontal regions brings up an interesting point concerning the calculation of mean quantities, such as velocity, from drifter data. Because of the nonuniform distribution of the buoys such quantities would be biased by the contribution of the high-speed currents associated with the frontal regions and, therefore, may not represent a true area average.

The mean near-surface speeds presented in Table 3 suggest that the SAF and PF are circumpolar in nature. *Nowlin and Clifford* [1982] show that at Drake Passage these two fronts account for approximately 60% of the total transport of the ACC. If this observation is representative of the general circulation of the ACC, then over half of the transport of the ACC would occur in two narrow high-speed currents.

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